



CS2016 – PROGRAMMING FOR BUSINESS
SEMESTER FINAL PROJECT
REAL WORLD BUSINESS PRODUCT
PROJECT REPORT

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INVENTRO ANALYTICS

Inventory simplified – Profits Amplified

Track Sales | Manage Stocks | Maximize Profits

THE README

Must read before running the program

FINAL PROJECT REPORT
INVENTRO ANALYTICS

Steps to follow before running the code:

If you are running the code for the very first time:

- The Code file will only work on JUPYTER ENVIRONMENT.
- Upload the notebook in specified folder, usually location goes into C drive > users.
- When you have uploaded the notebook in the specified folder, make sure to upload the excel file of dataset (Project DataSet.xlsx), in the same folder along with the notebook file.
- Then open the code file, there you will find all the modules separated in single cells.
- Run only modular functions' cells one by one, (avoid executing text boxes), until you reach the **12th Module (The Main Function)**.
- Now, you can enjoy the Inventro Analytics by running the 13th cell:
execute_product().
- The Product is password protected and passwords are mentioned in report as well as in the **password_authentication()**.

But if you do not run the code and just checking the file with its output on the end:

- Then you must keep in mind the following sales and customer records have been recorded over the time period of 2 days (3rd Dec and 4th Dec)

If you are running the code for the very first time:

The Sales Record – RUN 01 – **3rd December, 2025 – 1:15 PM**

Trans. ID	Product ID	Product Name	Quantity Sold	Category	Cust. ID
01	1	Tapal Danedar	6	Tea / Beverages	01
02	5	Lux Soap	10	Personal Care	04
03	43	Super Kernal Rice	10	Grocery / Staples	04
04	47	Surf Excel	15	Homecleaning	03
05	15	Kurkure	19	Snacks	02
06	232	Knorr Noodles	5	Kids Items	02
07	236	Gold Leaf	10	Cigarettes	01
08	200	Prince Biscuits	12	Biscuits / Bakery	05
09	218	Mosquito Coil	10	Household Items	06
10	242	Milkpak Milk	10	Dairy	04

The Customers Record - **3rd December, 2025 – 1:15 PM**

Cust. ID	Customer Name	Phone	Trans. ID
01	Moeez	1122334455	1, 7
02	Amna	123456	5, 6
03	Aimen	13579	4

04	Tooba	246810	2, 3, 10
05	Nayab	111222	8
06	Ayan	109876	9

The Sales Record – RUN 02 – 4th December, 2025 – 12:15 AM

Trans. ID	Product ID	Product Name	Quantity Sold	Category
11	377	Ashraffi Atta	1	Grocery / Staples
12	54	Lipton Yellow Label	10	Tea / Beverages
13	38	Bunny Bread	10	Biscuits
14	177	Yogurt Olpers	10	Dairy
15	111	Lays Chips Masala	17	Snacks
16	101	Areil	18	Homecleaning
17	65	Sunsilk Shampoo	10	Personal Care
18	93	Cerelac	10	Kids Items
19	62	Chamber	5	Cigarettes
20	3	Matchbox	1	Local household items

The Customers Record – No Customer Record Because all these transactions are done by Walk-In Customers

THE PROJECT REPORT

The Briefs and details of the product and its background

The Introduction

Stage 01

Introduction:

Project Overview:

Inventro Analytics is a Python-based real-life application designed to help retail stores manage their daily operations with efficiency and clarity. In today's competitive retail environment, shop owners struggle with manual record-keeping, inconsistent stock control, and lack of real-time insights. This project aims to automate and streamline the complete workflow, from tracking inventory and recording sales to analyzing profits and visualizing business performance.

Purpose & Significance:

The primary purpose of this system is to deliver a simple, reliable, and effective analytical tool tailored for small and medium-sized retail businesses that often cannot afford advanced Enterprise Resource Planning (ERP) solutions. Many local shop owners and small retailers struggle with issues such as unmanaged inventory, inconsistent sales tracking, stockouts, overstocking, and the lack of meaningful insights to support everyday decision-making. This application addresses these gaps by offering an accessible, Python-based solution that transforms raw sales and inventory data into actionable business intelligence.

The significance of this project lies in demonstrating how fundamental programming and data-processing concepts can be used to solve real-world business problems with measurable impact. It showcases how Python, through file handling, functions, control structures, and basic analytics can be leveraged to build a functional management tool that mirrors the essential features of costly ERP software but at a fraction of the complexity.

Problem Statement & Scenario:

Retail businesses, especially small and medium-sized stores, still depend heavily on outdated manual methods to run daily operations. These traditional systems create multiple operational and financial challenges:

- **Unclear Stock Information:**
Retailers cannot quickly view what is available, out of stock, or running low.
- **Manual Errors leading to stockouts or overstocking:**
Poor forecasting causes lost sales or wasted capital.
- **Difficulty Identifying Profitable Products:**
No structured system to track which items generate the most revenue.
- **Scattered Sales Data:**
Sales are noted in registers, slips, or verbally, making them hard to analyze.
- **No Automated Trend Analysis/Summaries:**
Retailers cannot monitor daily/weekly/monthly performance accurately.
- **No Visual Representation of Store's performance:**
Absence of charts or dashboards prevents understanding of growth patterns.
- **Inefficient Decision Making:**
The lack of organized data causes financial losses and operational delays.

The Real-World Example (New Wali Store, Ghalib Market, Gulberg):

To better understand the real-life challenges faced by retailers, consider a commonly visited general store in Lahore's area, Gulberg 3, Wali Store (Ghalib Market, Gulberg). It serves customers from almost three full blocks, offering a wide variety of daily-use products similar to those included in our dataset (snacks, beverages, household items, groceries, chocolates, noodles, etc.).

Despite handling **huge daily footfall** and a wide product range, the store still operates using manual systems.

Current Situation at Wali-Store:

- Sales are recorded manually in registers or on small slips.
- Stock levels are updated through guesswork, usually after the owner visually checks shelves.
- The store deals with hundreds of SKUs, but has no digital system to track them.
- Staff rely on memory to know which items are available or running out.
- The owner regularly faces:
 - Stockouts of fast-moving items
 - Overstocking of slow-moving items
 - Difficulty remembering which products are profitable
- Trouble calculating accurate profit margins
- During busy hours, there is:
 - No quick way to check inventory
 - No structured record of daily sales



Impact of Manual System at Wali Store:

- **Delayed Decision Making:**
Owner cannot quickly determine what to reorder.
- **Financial Losses:**
High chance of keeping too much stock or missing sales due to stockouts.
- **Scattered Data:**
Sales information is spread across slips, registers, and verbal notes.
- **No Performance Insights:**
The owner cannot see which categories bring the highest profit.
- **No Trend Tracking:**
Daily/weekly/monthly growth is impossible to analyze manually.

Why this scenario proves the need for Inventro Analytics?

Wali Store represents thousands of similar small and medium-sized retail shops across Lahore and Pakistan that:

- Handle a large variety of products
- Serve hundreds of customers
- But still rely on manual tracking

Inventro Analytics provides exactly the kind of centralized, automated, and intelligent system that can transform such stores from manual guesswork to data-driven operations.

The Product Features, Data Engineering and Ui/Ux

Stage 02

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INVENTRO ANALYTICS

The Core Functionalities:

Inventory Management:

Inventro Analytics has implemented and managed the inventory tracking and categorization module so efficiently that it includes all those common and important features that are must for retail operations:

- Add new products.
- Update quantities after restocking.
- Delete products from the inventory.
- Search for product details.
- View complete product list.
- Categorize products.
- Receive low-stock alerts when quantity falls below the reorder level.

Sales Management:

Inventro Analytics has a full support to deal with daily sales operations and whatever a common retailer wishes to do.

- Record sales with validation
- Automatically reduce inventory after each sale
- Prevent selling items with insufficient stock
- Generate daily, weekly, and monthly sales summaries
- Provide invoices and track customer purchases

Analytical Insights:

Inventro Analytics has a built-in dashboard component feature to help the retailers in decision making by providing them with the desired insights into their businesses through automated and not so complex visualizations:

- Calculate profit per product
- Identify top and least profitable products
- Monitor category-wise performance
- Detect negative profit margins
- Forecast profits using a simple moving average model

Visualizations:

It includes the following features:

- Bar charts for sales and profits
 - Line charts for trends over time.
 - Pie charts for category contributions
 - Histograms for distribution analyses
-

The Data Engineering:

The Data/File Handling:

Inventro Analytics has handled this part very carefully by linking OS with the product and allowing it to read, write and process the data sets smoothly, both in product and in local external storage.

- Defined core files for the application:
 - `products.csv` → stores all product-related data
 - `sales.csv` → stores all sales transactions
 - `customers.csv` → stores customer information and loyalty data
 - `Project DataSet.xlsx` → original dataset from which CSVs are generated
- Created a **backup directory** (backups) to store automatic backups of CSV files.
- Ensures all necessary directories exist before starting operations, avoiding runtime errors.

The Data Loading/Processing:

- **Products Data:**
 - Checks for existing CSV; if not found, converts Excel sheet into CSV for backend operations.
 - Ensures all product information is loaded and ready for inventory management.
- **Sales Data:**
 - Loads daily sales transactions from CSV.
 - Creates an empty Data Frame if file doesn't exist, ensuring system can track sales from day one.
 - Converts date fields to proper datetime format for accurate analysis.
- **Customers Data:**
 - Loads existing customer records; creates empty Data Frame if file doesn't exist.
 - Supports loyalty programs and customer tracking.
- Overall Ensures **data is ready, clean, and structured** for all system operations.

The Storage Handling (Save & Backup):

- **Automatic Backup:**
Saves timestamped backup copies of CSV files every time an update occurs.
- **Persistent Storage:**
Ensures all changes to products, sales, and customer data are safely written to the respective CSVs.
- **Centralized Management:**
Provides reliable data integrity while maintaining historical copies for recovery.

After every important operation, the updated dataset is automatically written back into the CSV file—forming a closed data loop that ensures accuracy and prevents data loss.

The structure of CSV File:

Inventro Analytics will only take in the excel file if organized the following way using following design structure in order to run functions smoothly.

Core Product Information

Attribute	Description
ProductID	Unique Identifier
ProductName	To display and identify the item
Category	Needed for filtering, summaries and category-wise sales etc.
Supplier	Will be used in restocking process for easier access

Inventory Details

Attribute	Description
CostPrice	For Profit Calculations
SellingPrice	For Sales and Profit Calculations
QuantityAvailable	Current Stock (update after each sale)
ReorderLevel	Will be used in the “low stock alert feature”

For Sales – Preprocessing columns must be included or they will be created manually during calculation or program running!

Attribute	Description
TotalValueInStock	$Q \times SP$ (will be calculated automatically)
ProfitPerUnit	$SP - CP$ (Auto calculation)
PotentialProfit	$Q \times \text{Profit per unit}$ (Will be used in forecasting)

Note for Retailers, Shopkeepers and Users:

- The data will be sourced from the retailers CSV files, which they will be needed to upload in order to use the product.
 - **Data Organization is the responsibility of the retailers**, shopkeepers and customers who will be using our product.
 - **Our product will only handle data manipulation, data sorting, data filtering and cleaning process.**
 - The Retailers, Shopkeepers and the customers using our product will need to store their raw data in excel file in the following described attributes format, so that our product processes it smoothly.
-

User Interface/User Interactions (UI):

Authentication & Access Control:

- Supports **three types of users** with fixed passwords:
- Manager → abc123
- Admin → efg456
- Analyst → xyz789
- Users have **role-based access**, ensuring security and proper workflow control.
- Login attempts limited to **3 tries**; system shuts down after maximum failed attempts.

Main Menu Navigation:

- Provides a **clear, color-coded command-line interface (CLI)** for easy navigation.
- Main menu options include:
 - Inventory Management
 - Record Sale
 - Profit Analyser
 - Visualizations
 - Generate Daily Report
 - Exit

Inventory Management:

Users can:

- View all products in a tabular format.
- Search products by **ID or Name** using smart search.
- Add new products with all required details (price, quantity, category, supplier, reorder level).
- Update stock quantities; automatically recalculates potential profit and total value.
- Delete products when no longer sold.
- Receive low stock alerts for items below 10 units.

Sales Recording:

- Supports **dynamic cart creation**: scan/search product, input quantity, calculate total & profit.
- Updates **inventory in real-time** after each sale.
- Tracks customer purchases with optional **loyalty program**:
 - Adds new customers automatically.
 - Updates points based on total profit of sale (10 points per 100 Rs).
- Generates a **beautiful text-based invoice** including:
 - Itemized list, quantity, price, total amount.

- Total profit and customer loyalty points.
- Creative profit meter visualization for quick understanding.

Profit Analysis:

Provides insights including:

- Total profit to date.
- Top 5 profitable products.
- Profit distribution by category.
- Moving average & short-term profit forecast.
- Potential profit from remaining stock.

Visualizations:

Users can generate **interactive charts**:

- Daily sales trend (line chart with value labels).
- Profit by category (pie chart).
- Top 10 products by profit (horizontal bar chart).

Report Generation:

Generate **daily summary reports** saved as text files including:

- Total sales, total profit, items sold.
- Low stock alerts.
- Top 5 products of the day.

User Experience (UX) Enhancements:

- Color-coded CLI for clarity and aesthetics.
 - Transposed tabular views for **full product details**.
 - Easy navigation with clear prompts and informative success/error messages.
 - Creative features like ASCII profit meter for visual feedback.
-

The Implementation of Key Programming Concepts

Stage 03

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INVENTRO ANALYTICS

Key Programming Concepts Implemented:

OOP Claas:

- Defined a Color class to store CLI Color codes for better readability and aesthetics.
- Encapsulation of Color codes ensures **clean and reusable UI styling**.

File Handling:

- Used Python's `os` and `shutil` modules to manage files and directories.
- Automatically creates a **backup folder** for safety.
- Reads from CSV or Excel depending on availability (`products.csv`, `sales.csv`, `customers.csv`).
- Updates and backs up files after each **critical operation** (sale, stock update, product addition).

Data Structures:

- **Data Frames (from pandas)** are used for storing and manipulating products, sales, and customer data efficiently.
- Lists are used to manage the **shopping cart** dynamically during sales.
- Dictionaries store individual product/sale/customer details for quick operations.

Conditional Statements:

- Role-based access using **if-elif-else** for manager, admin, and analyst.
- Low stock alerts, invalid input handling, and login attempt tracking implemented via conditions.

Loops:

- **While loops** for menu navigation and iterative input until user exits.
- Iteration over Data Frame rows for updating inventory, recording sales, and calculating loyalty points.

Functions:

- Modular design using **functions** for:
 - `load_data()` → Load CSV/Excel data.
 - `backup_and_save()` → Automated backup and saving.
 - `inventory_menu()` → Manage inventory operations.
 - `record_sale()` → Process sales with dynamic cart and loyalty system.
 - `profit_analyser()` → Compute total, top products, and category profits.
 - `visualizations()` → Plot interactive charts.
 - `generate_daily_report()` → Save daily summary report.

- Functions improve **code readability, reusability, and maintainability**.

Exception Handling:

- Used **try-except blocks** to handle invalid user input (quantity, ProductID selection).
- Prevents crashes and ensures smooth user experience.

Date and Time Handling:

Python's date time module used to:

- Track sales timestamp.
- Generate daily reports and forecasts.
- Maintain consistent record-keeping for profit calculations and visualizations.

Data Analysis & Manipulation:

Pandas operations for:

- Grouping (`groupby`) and aggregating sales and profit data.
- Sorting and filtering top products and low stock items.
- Calculating moving averages for **profit trends**.

Data Visualization:

- Matplotlib used for:
 - Line charts (daily sales trends).
 - Pie charts (profit by category).
 - Bar charts (top 10 products by profit).
- Enhances **data-driven decision-making** with visual insights.

Security Concepts:

- Password authentication with **role-based login**.
- Limited login attempts with enforced shutdown after maximum failures.

Modular & Scalable Design:

- Project structured in **modular functions**.
 - Easy to extend: adding new features, products, or reporting functions is straightforward.
 - Separation of **data handling, UI, analytics, and visualization** for clarity.
-

The Creative Feature of the Product

Stage 04

Creative feature of Inventro Analytics:

The Profit Analyzer:

Automated Profit Analyzer, helps retail stores enhance decision-making by providing actionable insights on store performance.

- **Profit Metrics:**

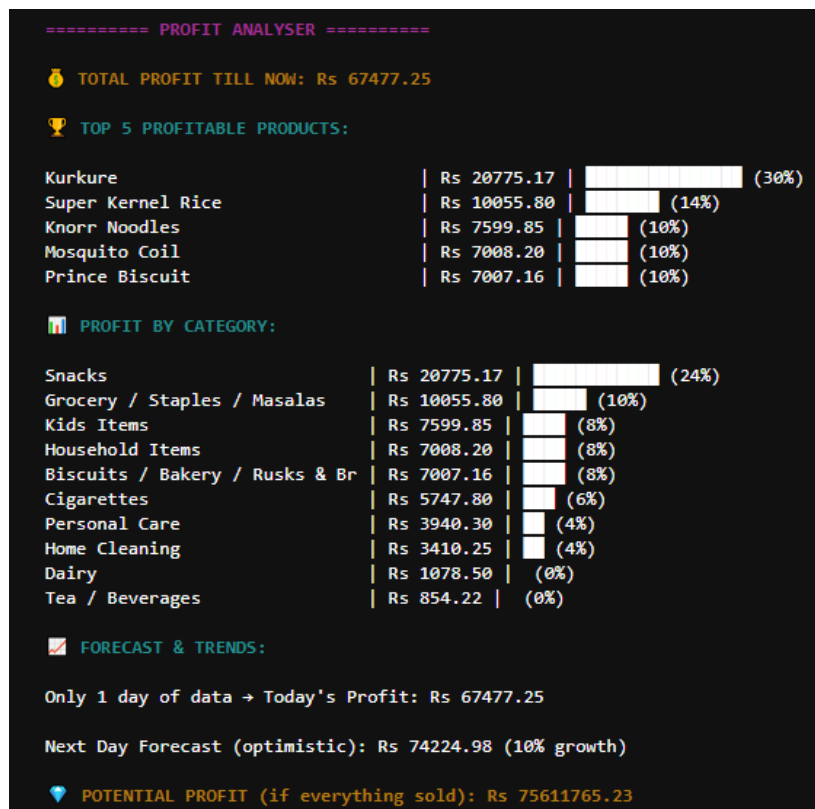
- Profit per product
- Profit per category
- Total profit
- Top 3 most profitable products
- Most profitable category
- Least profitable product & category
- Loss detection (negative profit margin)
- Profit forecasts using **moving average method**

- **User Interface / Console Visualizations:**

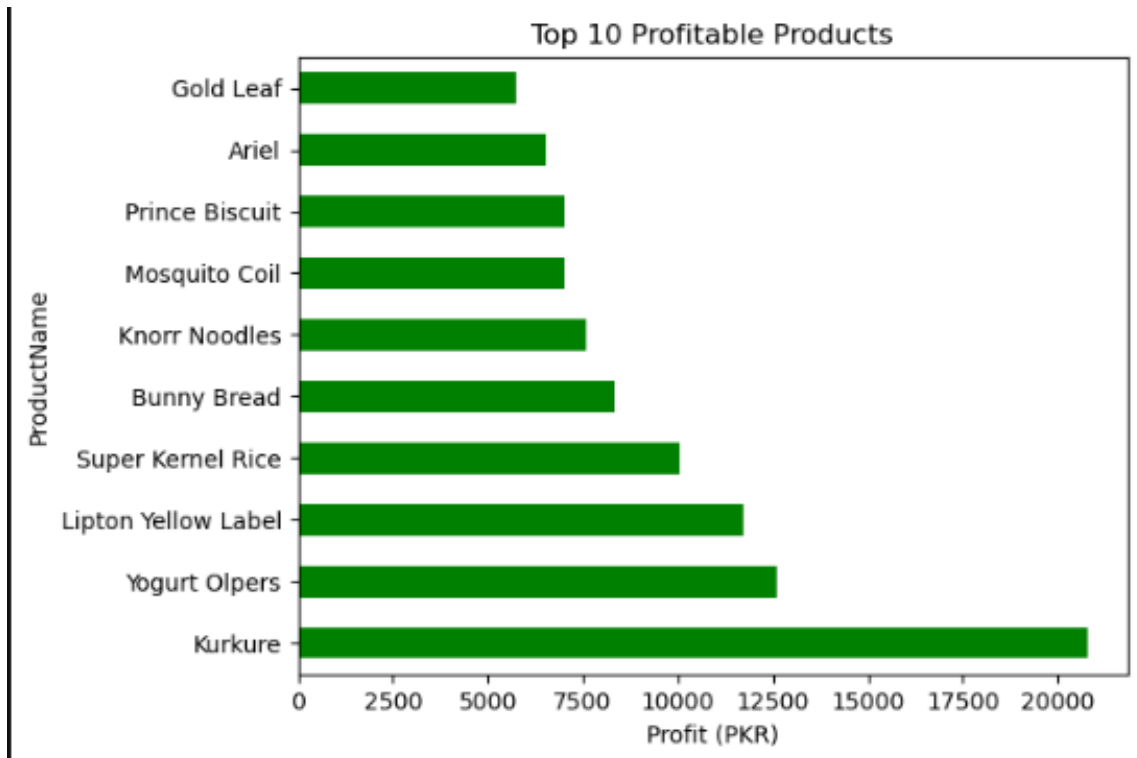
- **Bar Charts:** Profit per product (top 10 products)
- **Histograms:** Profit distribution per category
- **Pie Charts:** Contribution of each category to total profit
- **Line Charts:** Daily/weekly/monthly profit trends
- **Text Summaries:** Key statistics for quick decisions
- **ASCII Profit Meter:** Visual gauge of profit per sale

A little Demonstration:

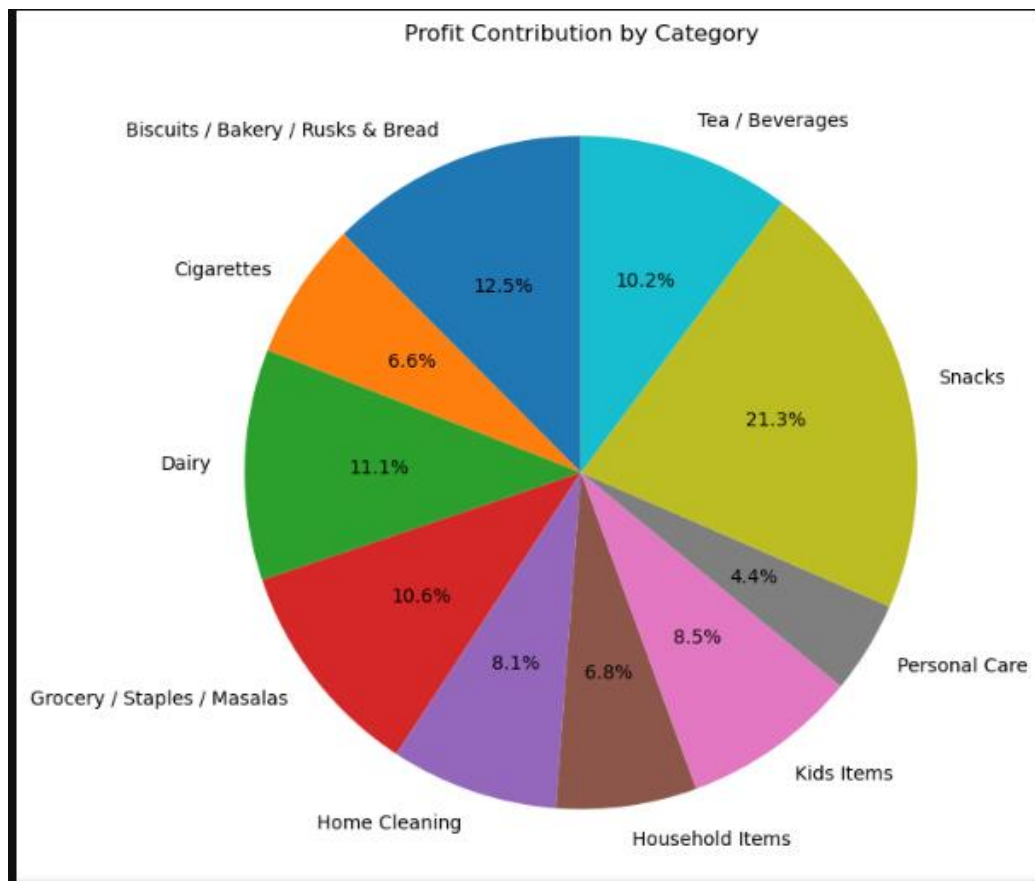
Automated Profit Analyzer Dashboard



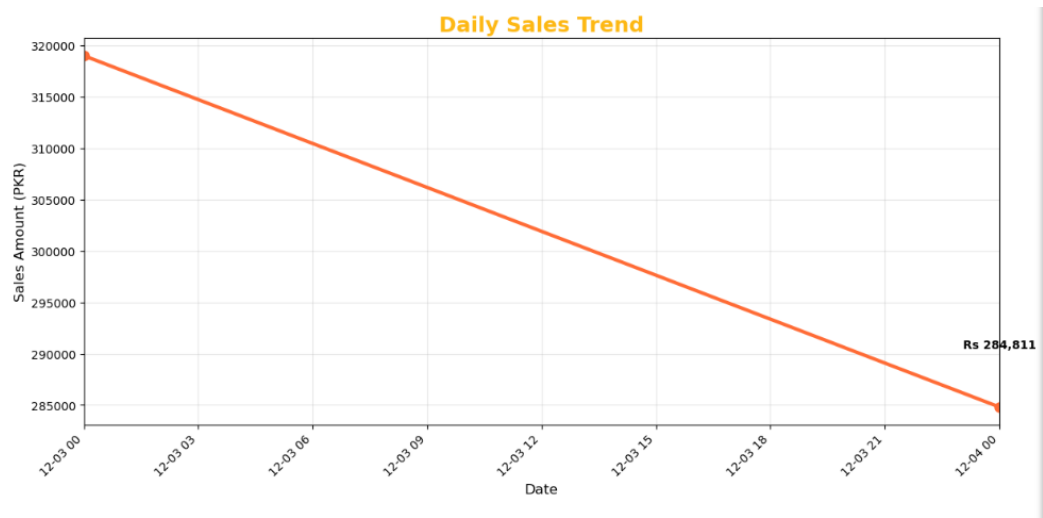
Automated Profit Analyzer Visualizer, Chart 01



Automated Profit Analyzer Visualizer, Chart 02



Automated Profit Analyzer Visualizer, Chart 03



Challenges, Summaries and Conclusion

Stage 05

Challenges Faced & Solutions Implemented:

The Challenges:

- File path errors and permissions issues.
- Data inconsistency and missing columns in user-provided CSV files.
- Unexpected crashes due to invalid user input.
- Difficulty managing frequently updated datasets.
- Handling multiple users with different access roles.
- Dynamic visualization of sales/profit data.

The Solutions:

- Used flexible path handling, relative file paths, and folder-existence checks.
 - Implemented preprocessing that auto-creates missing columns with default values.
 - Added extensive input validation and exception handling.
 - Adopted a closed-loop approach where every update is immediately written back to CSV.
 - Implemented secure login with role-based access and attempt limits.
 - Added conditional checks before generating charts and reports; used clear, readable visualization styles.
-

Summary & Improvements

The Summary:

Inventro Analytics successfully demonstrates how Python can solve real-world business problems faced by local retailers. It automates inventory management, streamlines sales tracking, generates meaningful visual insights, and provides a powerful profit analysis engine, all within a single integrated system. The project proves how core programming logic, data handling, and automation can be applied to build practical, impactful business applications.

Project Outcomes:

- Fully functional inventory management system
- Automated sales logging and instant stock updates
- Accurate profit calculations per product and category
- Visual insights through charts and summary reports
- Closed-loop data handling with auto-backup and storage
- Role-based system with clean, interactive console UI

Future Improvements:

- GUI version using Tkinter / Flask for real-world usability
- Shift from CSV to databases (SQLite / MySQL)
- Barcode scanner integration for rapid billing
- Multi-user interfaces and access levels
- Cloud dashboards for remote monitoring
- SMS/WhatsApp auto-alerts for low stock
- Machine learning integration for profit forecasting

The Conclusion:

Inventro Analytics stands as a complete and functional retail-management solution built entirely with Python. Through its integrated modules: inventory control, sales tracking, data automation, visualization, and profit analysis, the system addresses the real operational challenges faced by small and medium-sized stores. By replacing manual registers and guesswork with structured data and automated analytics, the application shows how even simple programming tools can create powerful, real-world impact.

The project not only fulfills academic requirements but also demonstrates the practical value of core programming concepts such as modular functions, data structures, file handling, validation, and automation. Its creative feature: the Profit Analyzer, further elevates the system by offering intelligence, forecasting, and decision-support capabilities that are rarely available to local retailers.

Overall, Inventro Analytics proves that well-designed code can transform traditional retail operations into data-driven, efficient, and insightful workflows. With further enhancements such as GUI development, database integration, and cloud-based dashboards, the system holds strong potential for real-world deployment and future scalability.
